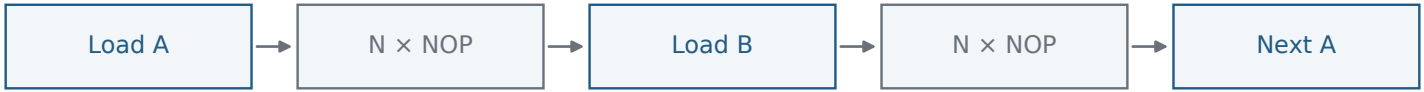
1 Workload and dependencies



Independent: A and B chase disjoint shuffled rings; each chain depends on its own prior load.
Dependent control: all loads follow one serialized cycle through the same complete line set.

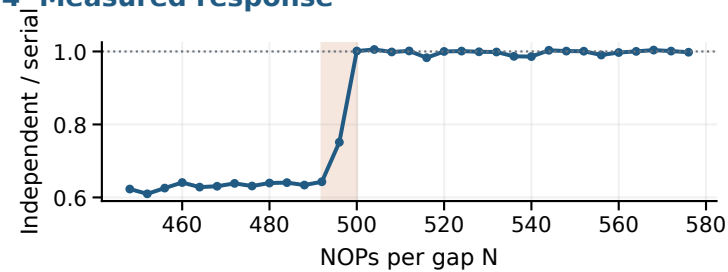
2 Controlled comparison

256 MiB random rings; 1-byte NOPs; fine N: Charnwood 192–240, Artemisia 448–576.

3 Timed batch and normalization

I=100,000 loops; 2 loads/loop; ticks/load = elapsed TSC / (2 I).

4 Measured response



5 Inference and control

Independent loads lose overlap as spacing grows and approach the dependent-load control. Repeated fine steps bracket an effective instruction window. N is per gap; neither N nor 2N is automatically a total ROB entry count.